Nuclear envelope: The nucleus is separated from the cytoplasm by a double membrane, the nuclear envelope and the two membranes separated from each other by a perinuclear space of varying width. There are little holes in the nuclear envelope called nuclear pores which help the substances to move into or out of the nucleus. DNA occupies most of the space inside a nucleus. DNA is the genetic material and provides the instructions essential for building proteins. Proteins are responsible for helping with most activities in a cell. Inside the nucleus is a round body called nucleolus, which is present in a eukaryotic cell. The nucleolus is devoid of an encircling membrane. The nucleolus produces the ribosomal subunits from proteins and ribosomal RNA, also known as rRNA. It then sends the subunits out to the rest of the cell where they combine into complete ribosomes. Ribosomes make proteins; therefore, the nucleolus plays a vital role in making proteins in the cell.

1.4.4.Mitochondria- the power houses of the cell

A cell has a compartment for energy production. It obtains energy from the food supplied by its environment. This energy then has to be converted into some form that can be distributed throughout the cell. The common solution is to store energy in the form of a common molecule that can be used whenever and wherever it is needed in the cell. The term 'mitochondrion' is derived from the Greek word 'mitos' which means 'thread' and 'chondrion' which means 'granule'. Mitochondria is a membrane bound cellular structure and is found in most of the eukaryotic aerobic cells. Mitochondria may assume different shapes ranging from granular to filamentous depending upon the functional state of the cell. They are spherical in yeast cells, elliptical in kidney cells, elongated in liver cells and filamentous in fibroblasts. The size of the mitochondria ranges from 0.5 to 1.0 μm in diameter.

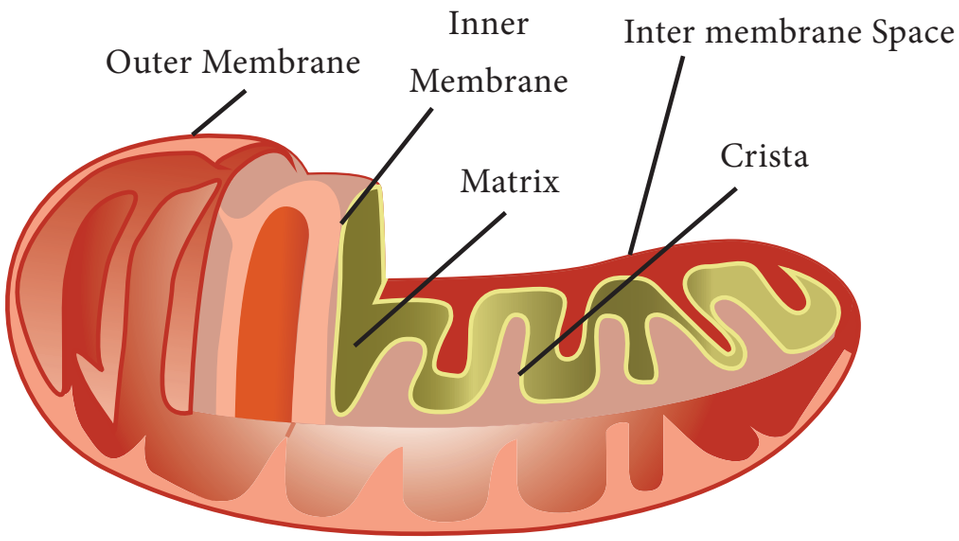


Figure 1.7 A schematic diagram of a mitochondrion

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